



Machine Learning

Decision Tree • Random Forest • SVM



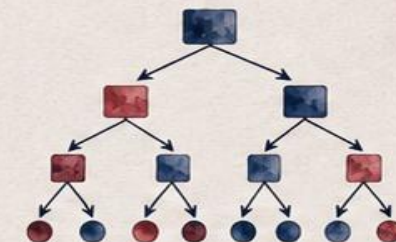
Presenters:
Negin Sharifi
Kian Mamdoohi



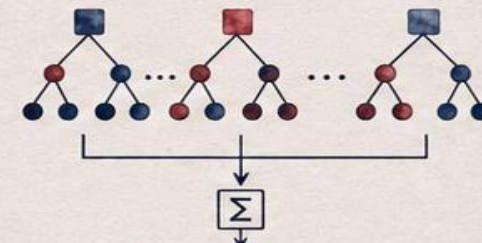
Instructor:
Dr. Ebrahimpour



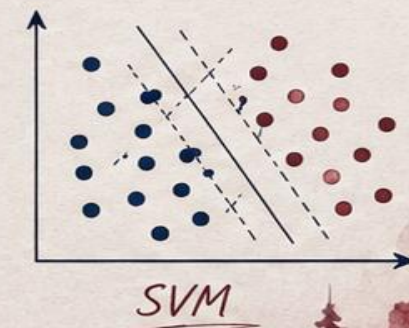
Date:
6 Jun 2026



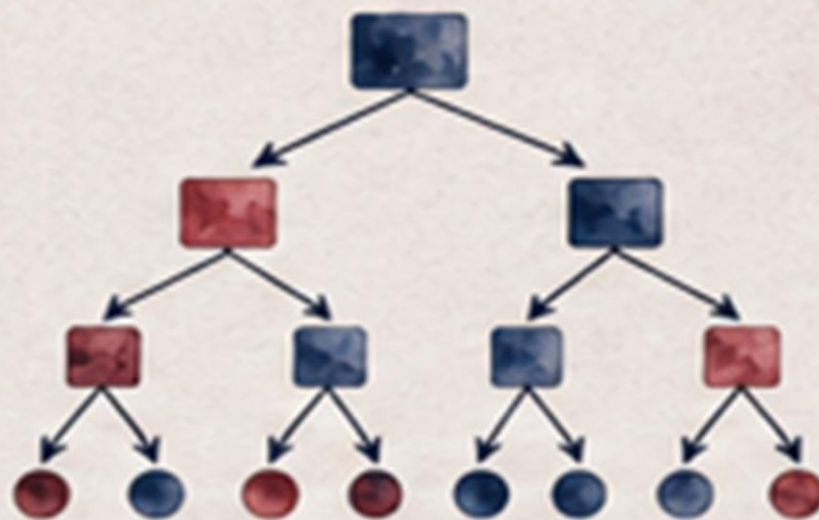
Decision Tree



Random Forest



1



Decision Tree



- ❑ Have you ever thought about how machines make decisions?
- ❑ Can machines really make decisions, or are they just following a set of simple rules?
- ❑ What exactly happens when a machine makes decision?





Do you have a cough?

Are you experiencing body aches?

Do you have a runny nose?

Do you have a fever?

Do you have a sore throat?





1 Decision Trees

Is there a guarantor?

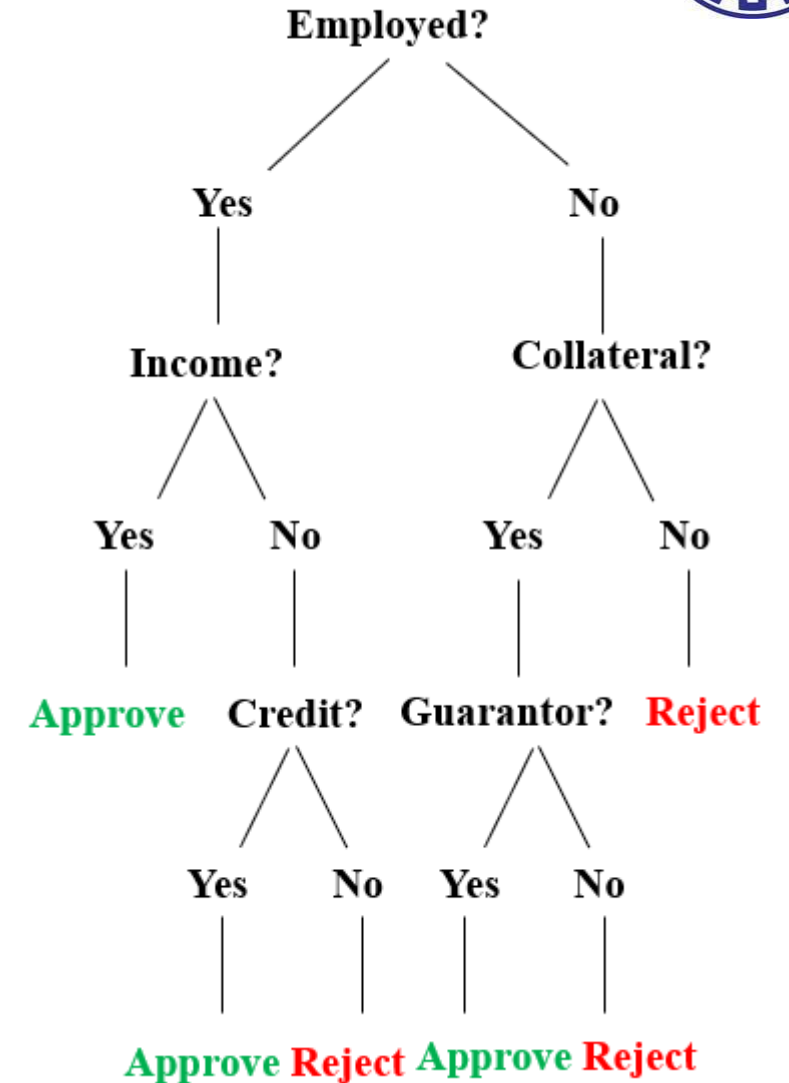
Is the applicant employed?

Does the applicant have a good credit score?

Is the income higher than a threshold?

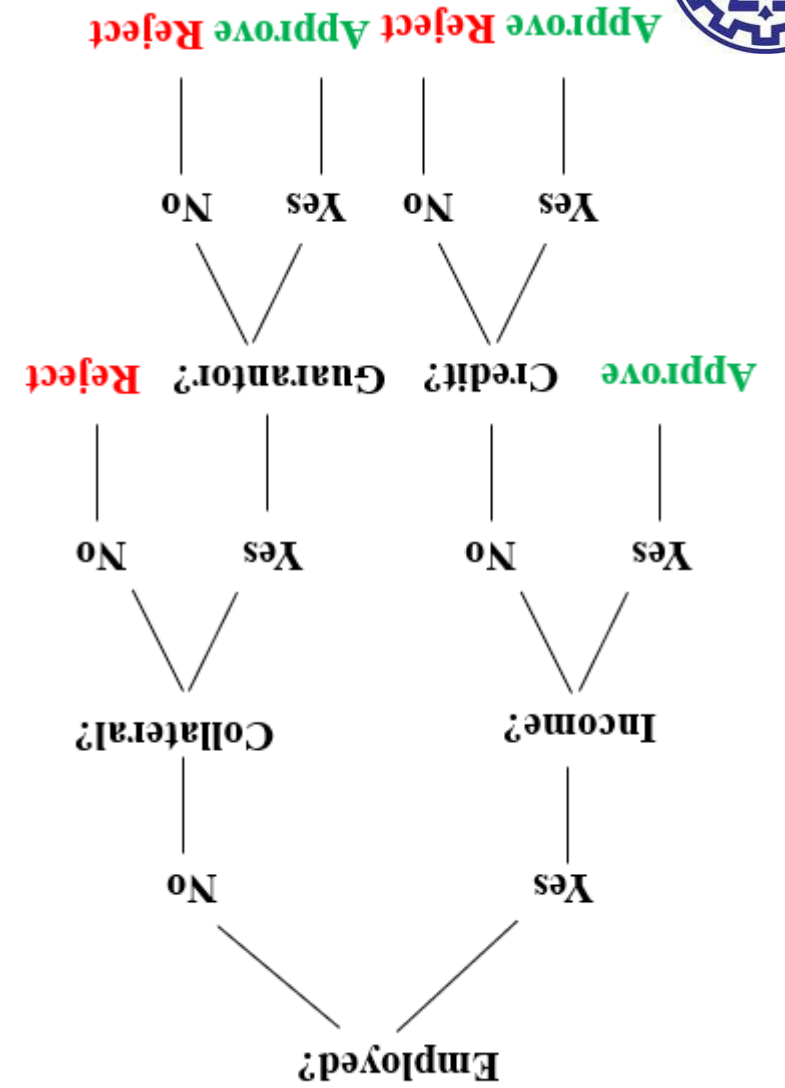
Does applicant have collateral?

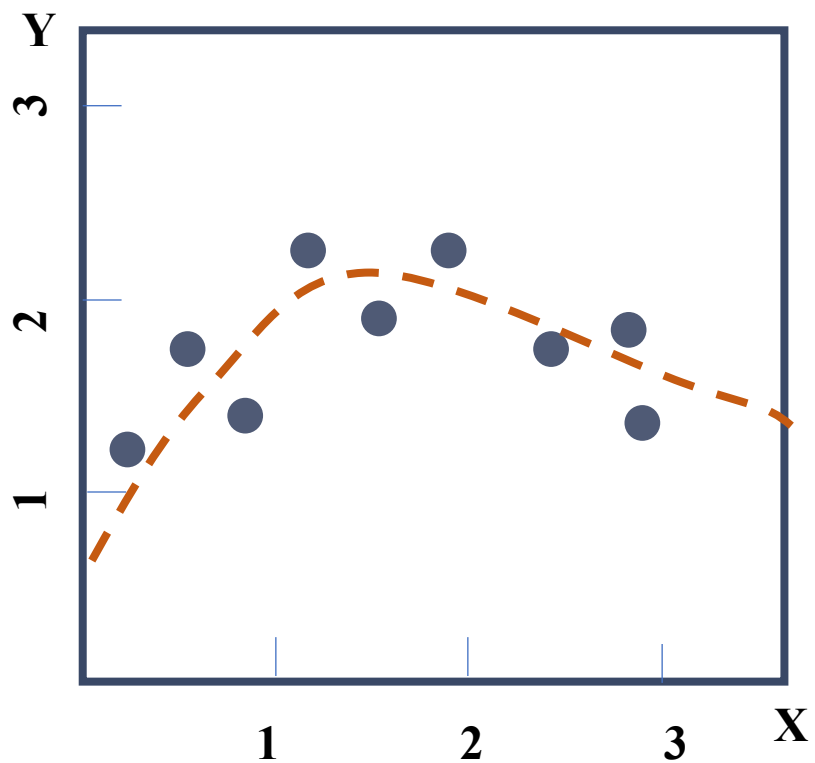
Yes!



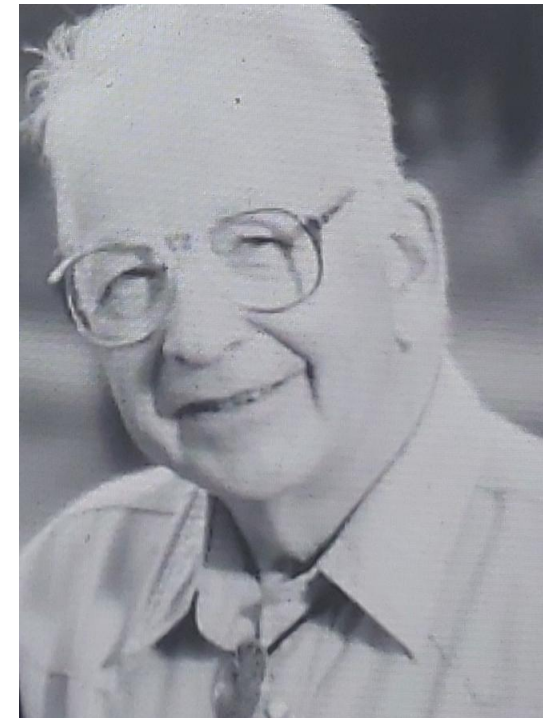
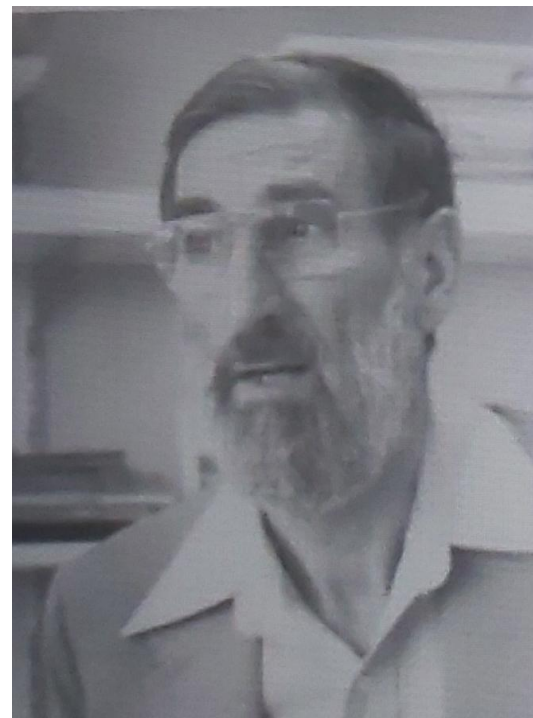


1 Decision Trees

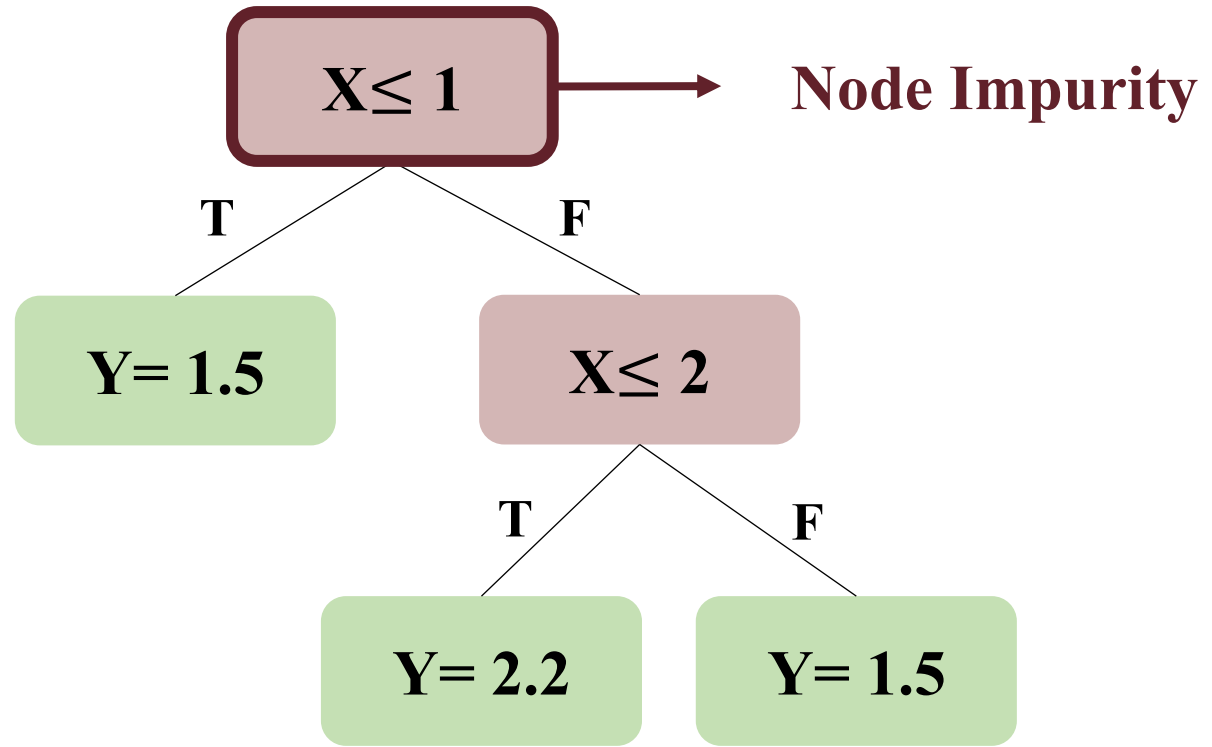
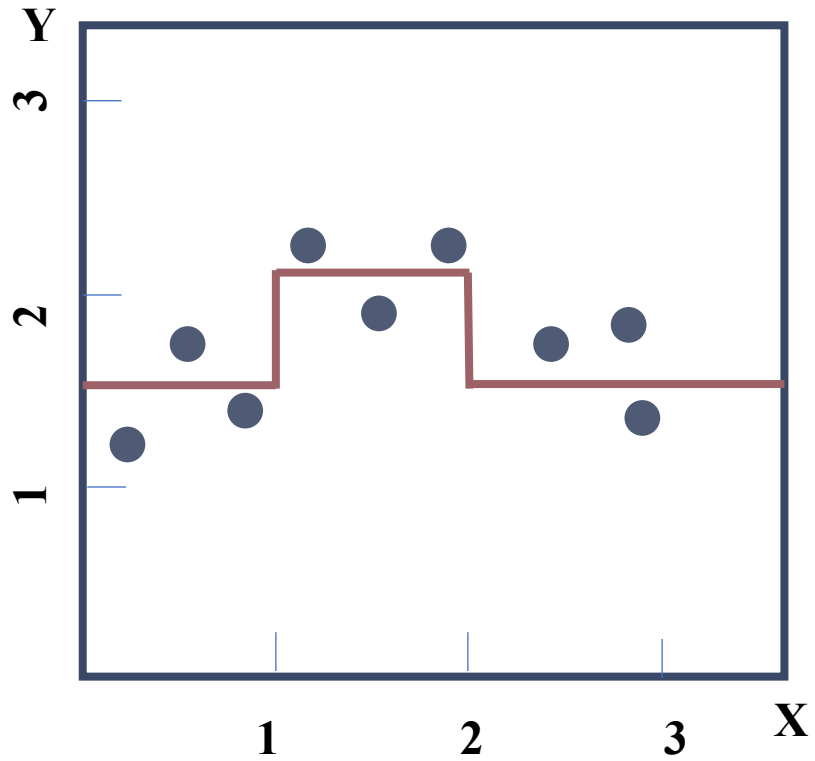




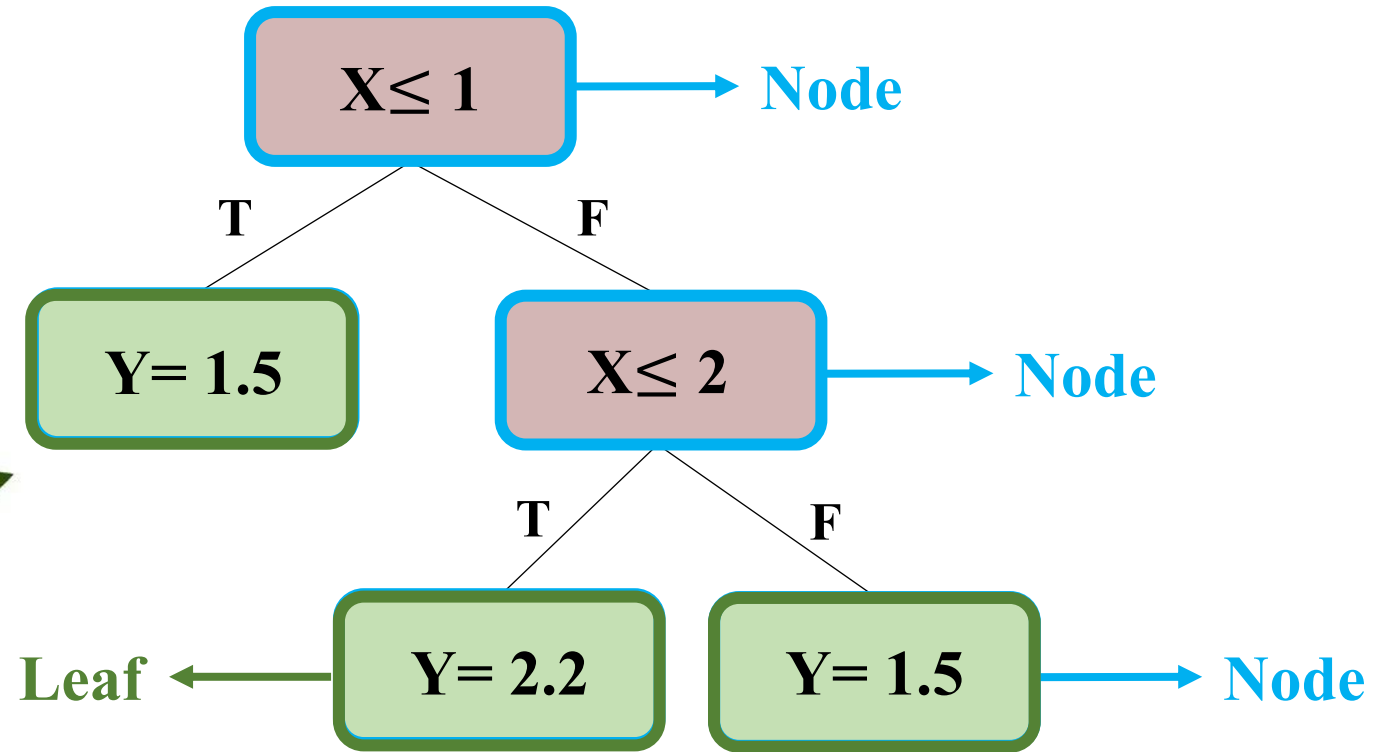
1963



1 Decision Trees



1 Decision Trees



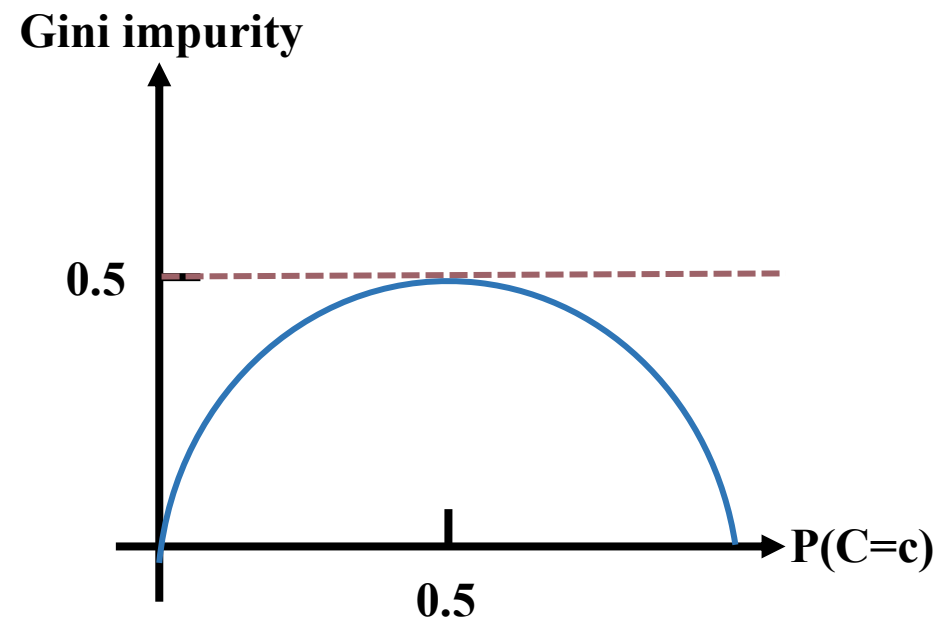


Gini Impurity:

Gini Impurity measures the degree of class impurity in a dataset.

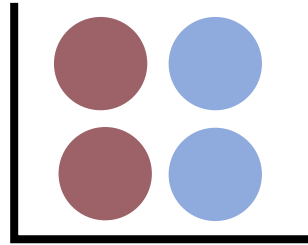
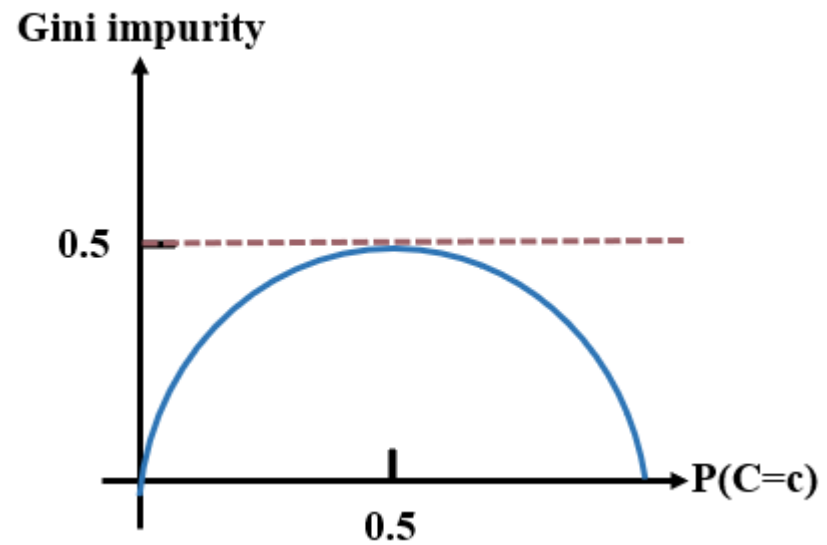
$$G(Q) = \sum_{c \in C} P_c (1 - P_c)$$

$$P_c = \frac{1}{N_Q} \sum_{x \in Q} \mathbf{1}(y_{\text{class}} = c)$$

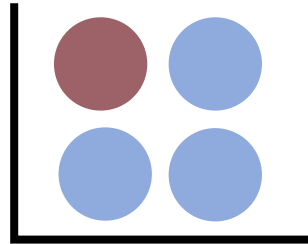




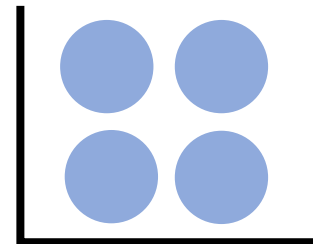
1 Decision Trees



Gini impurity= 0.5



Gini impurity= 0.375



Gini impurity= 0

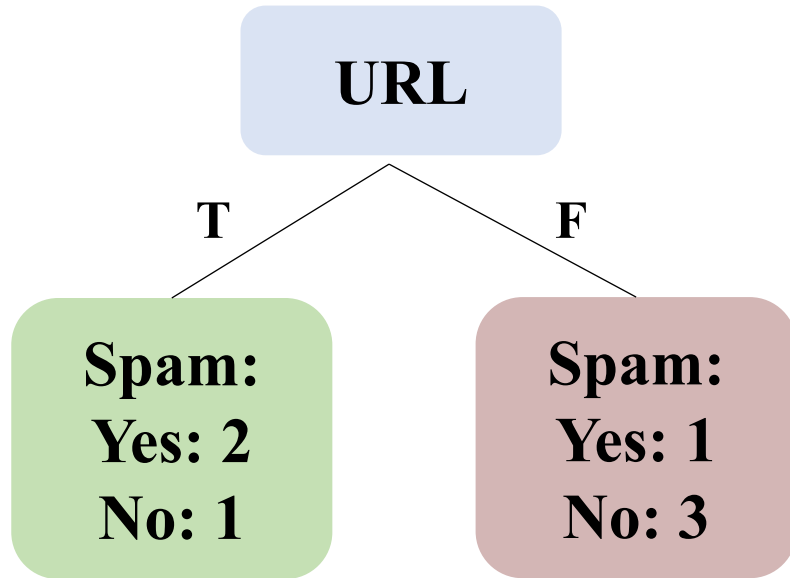


Decision Tree Construction Based on Gini Impurity:

X- URL Link	Y- Spam
Yes	Yes
Yes	Yes
No	No
No	No
No	Yes
No	No
Yes	No



Decision Tree Construction Based on Gini Impurity:



X- URL Link	Y- Spam
Yes	Yes
Yes	Yes
No	No
No	No
No	Yes
No	No
Yes	No



Decision Tree Construction Based on Gini Impurity:

X- Words in Email	Y- Spam
10	Yes
40	No
20	Yes
50	No
30	No



Decision Tree Construction Based on Gini Impurity:

X- Words in Email	Y- Spam
10	Yes
20	Yes
30	No
40	No
50	No

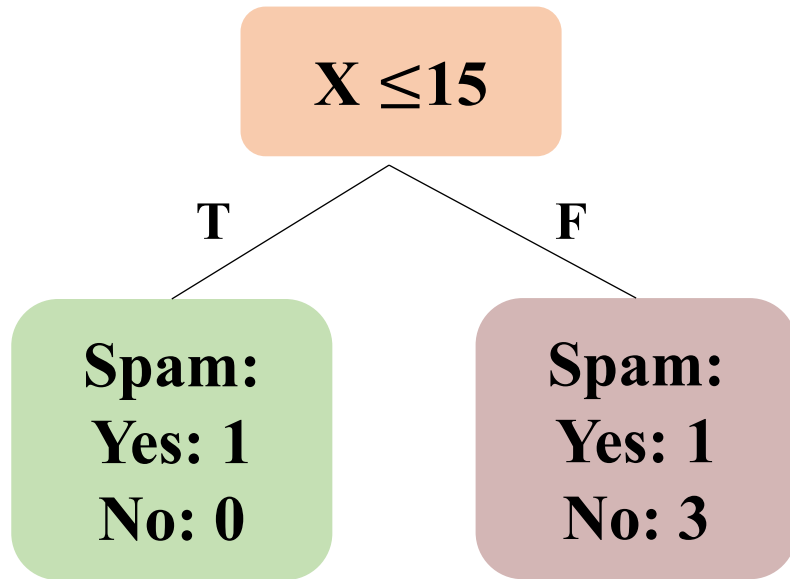


Decision Tree Construction Based on Gini Impurity:

X- Words in Email		Y- Spam
10	15	Yes
20	25	Yes
30	35	No
40	45	No
50		No



Decision Tree Construction Based on Gini Impurity:



X- Words in Email	Y- Spam
10	Yes
15	Yes
20	Yes
30	No
40	No
50	No

$$G(Q) = (1/5)(0+0) + (4/5)[(1/4)(1-1/4) + (3/4)(1-3/4)] = 0.3$$



Decision Tree Construction Based on Gini Impurity:

	X- Words in Email	Y- Spam
	10	Yes
	20	Yes
	30	No
	40	No
	50	No

Gini= 0.3	←	15
Min. Gini= 0	←	25
Gini= 0.26	←	35
Gini= 0.4	←	45

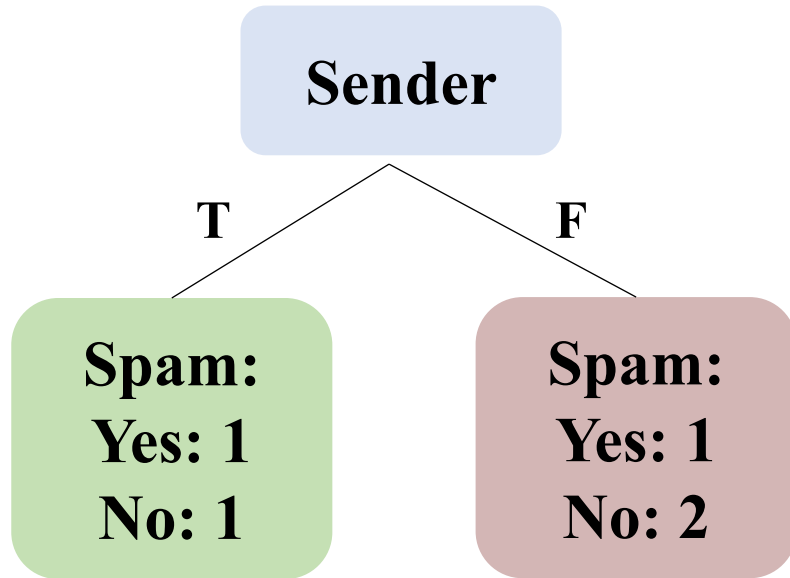


Decision Tree Construction Based on Gini Impurity:

X- Sender	Y- Spam
Abe	Yes
Bob	Yes
Claire	No
Abe	No
Bob	No



Decision Tree Construction Based on Gini Impurity:



X- Sender	Y- Spam
Abe	Yes
Bob	Yes
Claire	No
Abe	No
Bob	No



Decision Tree Construction Based on Gini Impurity:

X- Sender	URL	Words	Spam
Abe	Yes	10	Yes
Bob	Yes	20	No
Claire	No	30	No
Abe	No	40	Yes
Bob	No	50	No

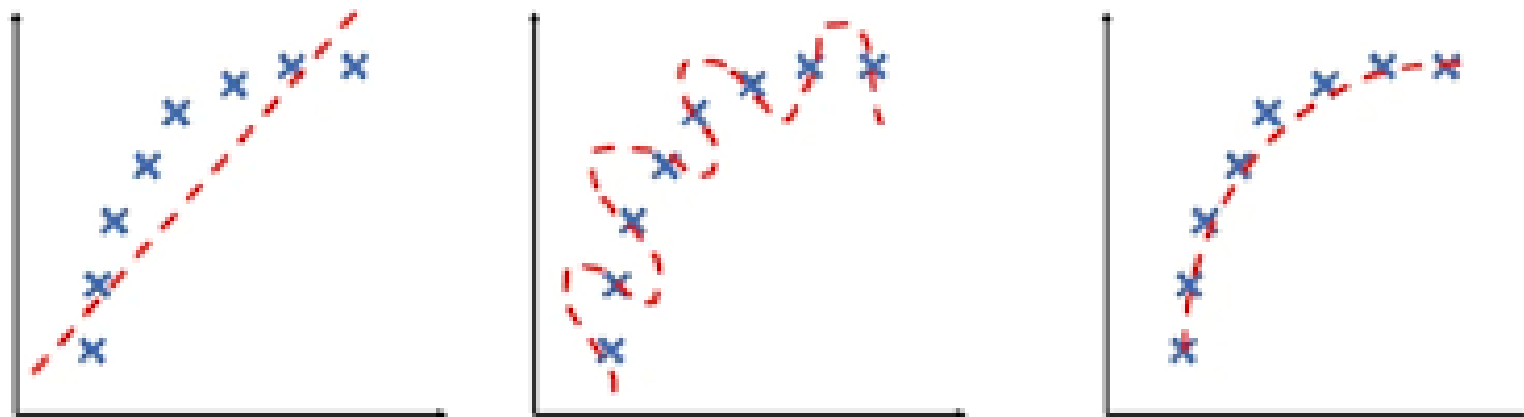


What can cause a machine learning model to perform poorly on new data?

OR

Why do some machine learning models perform very well on training data but poorly on unseen data?

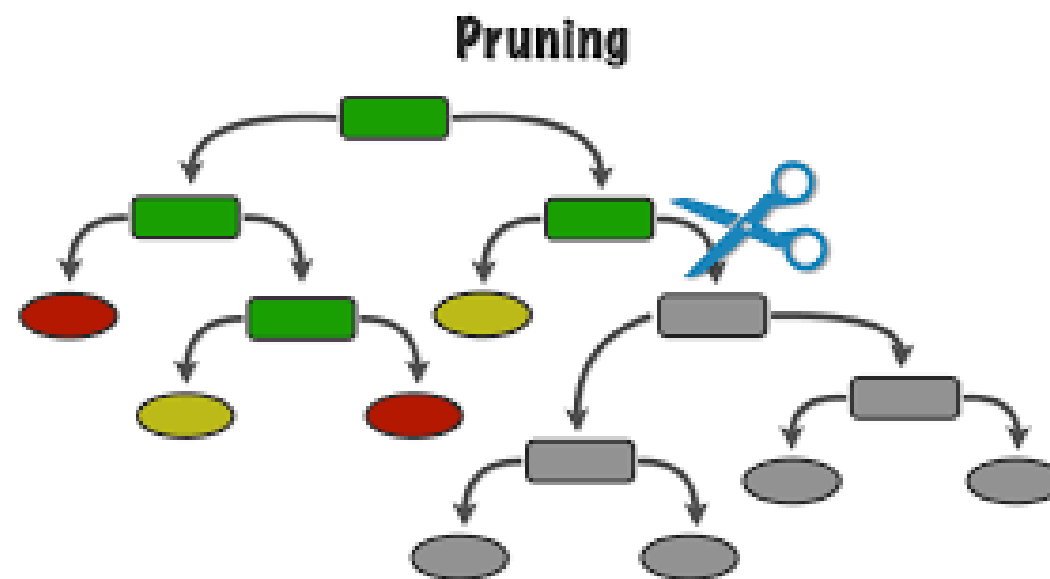
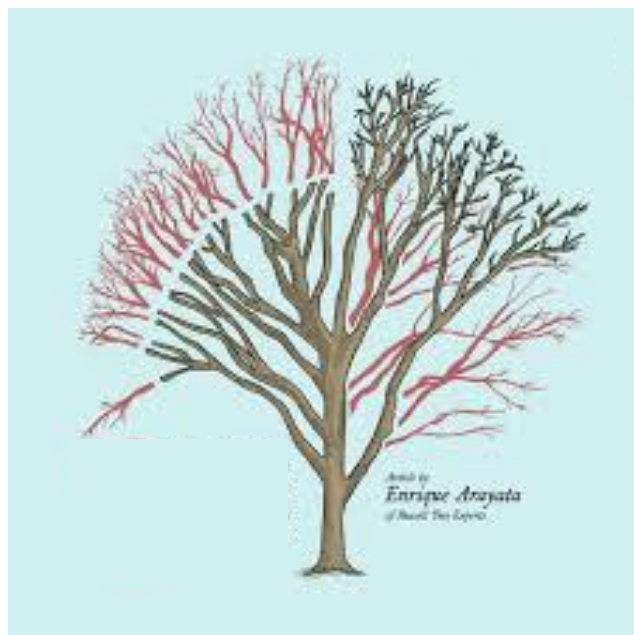
Overfitting!



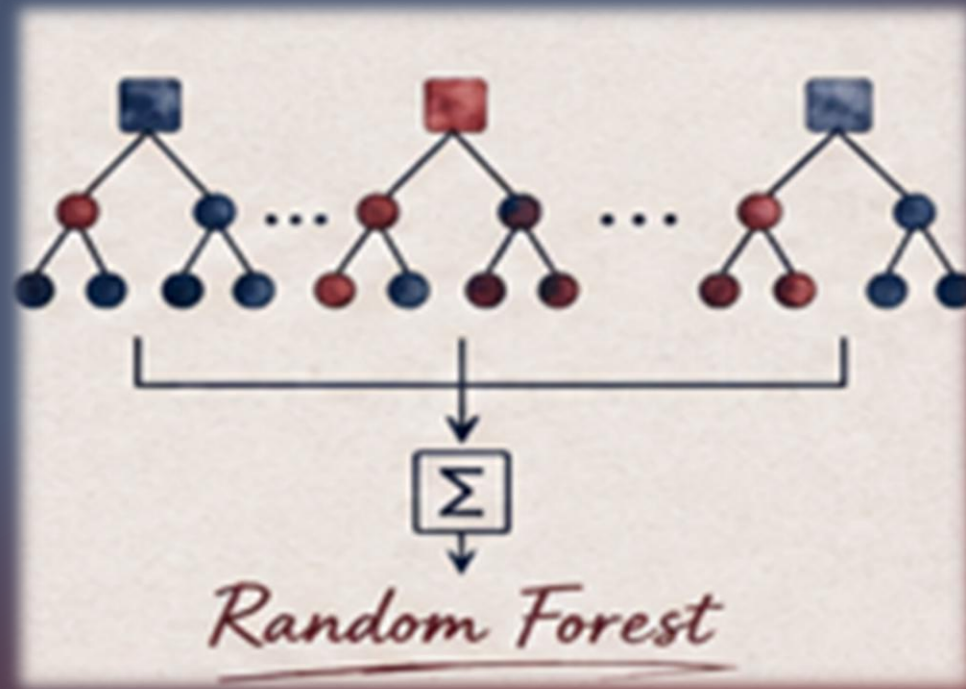
Decision Trees tend to grow very large and are prone to **overfitting**!

Pruning is used to prevent overfitting. There are several pruning techniques:

- Setting a minimum Gini impurity threshold of 0.01.
- Limiting the maximum tree depth to 3.



2





Problems of decision Trees:

- ❌ When using Gini impurity, there is no guarantee that all features will be utilized in the tree.
- ❌ With a fixed splitting criterion, the root node will always be the same.
- ❌ Decision trees are prone to overfitting.



Random Forests

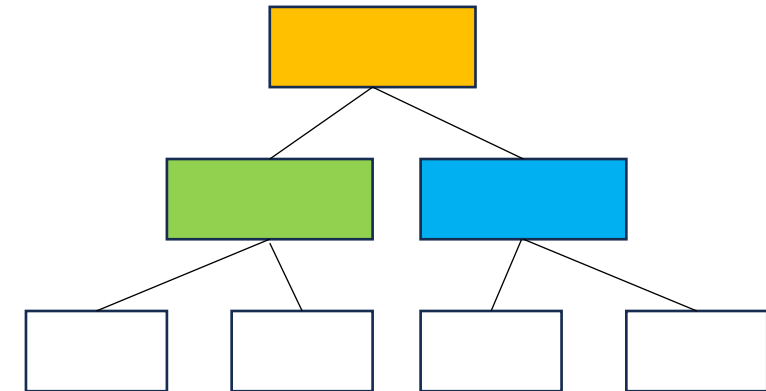




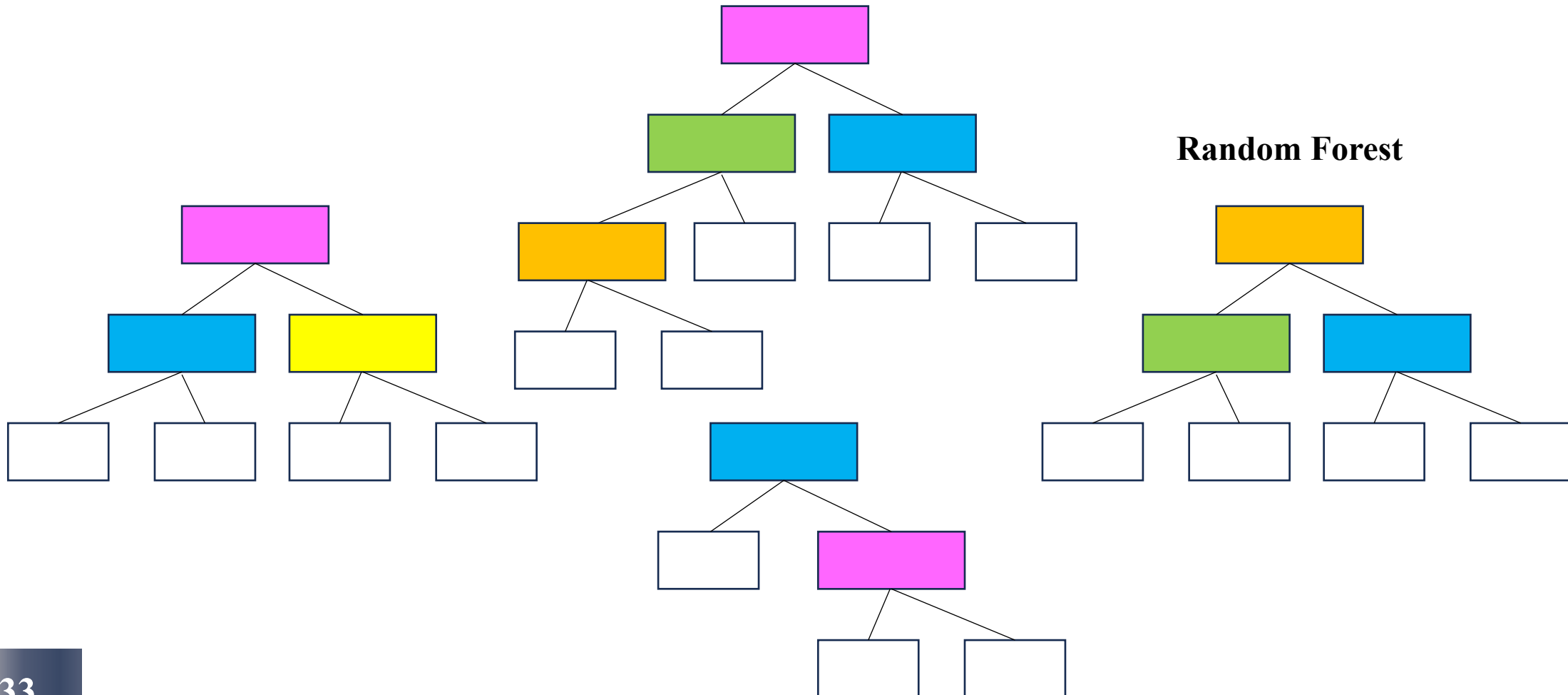
It was proposed that a random subset of features be selected at each split.

					Y

Decision Tree

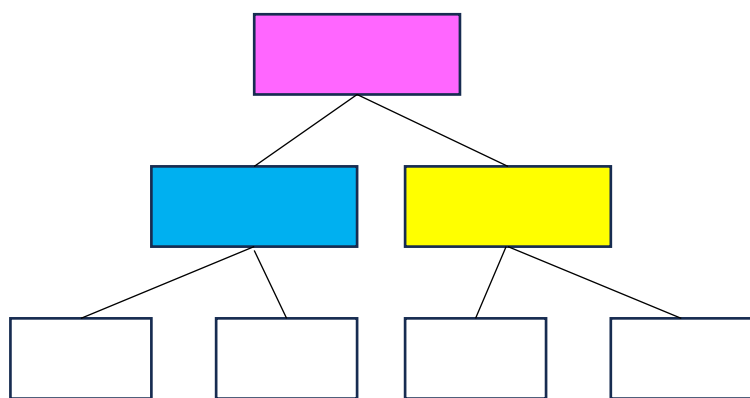


It was proposed that a random subset of features be selected at each split.

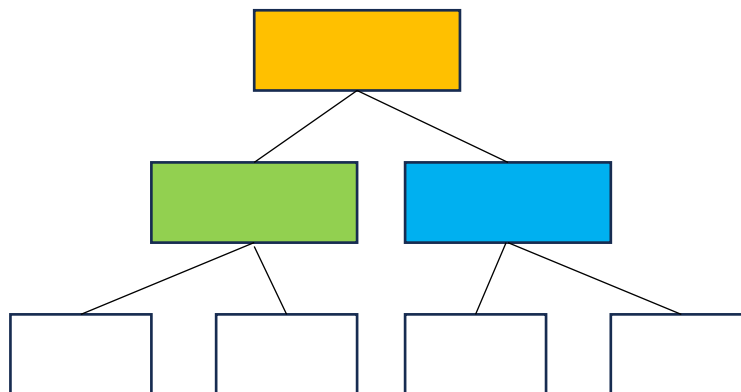




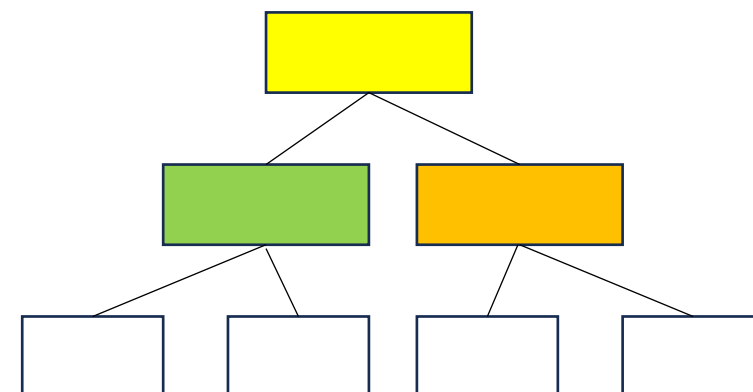
It was proposed that a random subset of features be selected at each split.



Tree 1: $y=0$



Tree 2: $y=0$



Tree 3: $y=1$

$y=0$

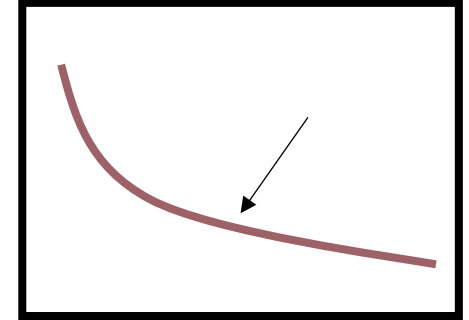


1. Number of Estimators:

Is there an upper limit to the number of trees?

Does increasing the number of trees lead to more overfitting?

Error



Number of trees

“Unlike a single decision tree, a random forest is generally not prone to overfitting. The number of trees can be increased as needed, and the method remains relatively fast and efficient!”



2. Number of Features:

How many Features should be used in each subset?

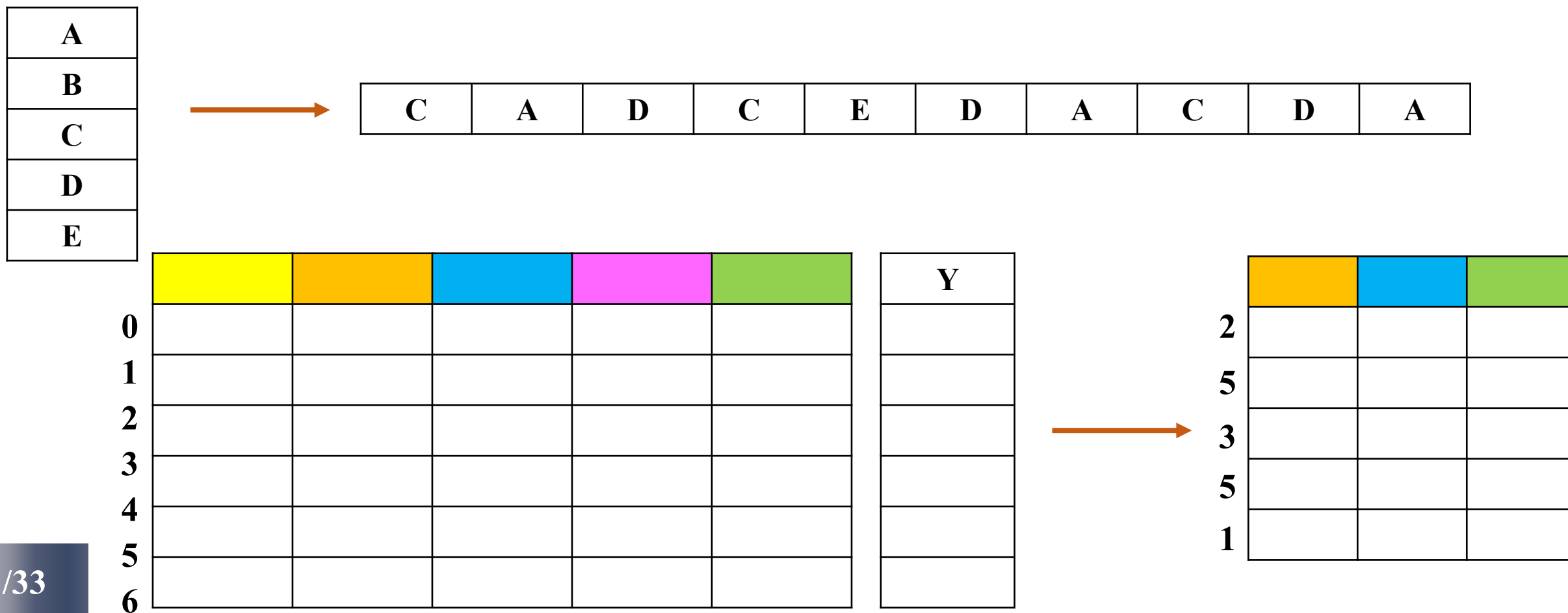
“There is an interesting difference between regression and classification: in both cases, correlation increases slowly as the number of features used increases.”

It has been suggested to use $\log_2(N+1)$, $\sqrt{N}$ or $N/3$ to determine the number of features.



3. Bootstrap Samples:

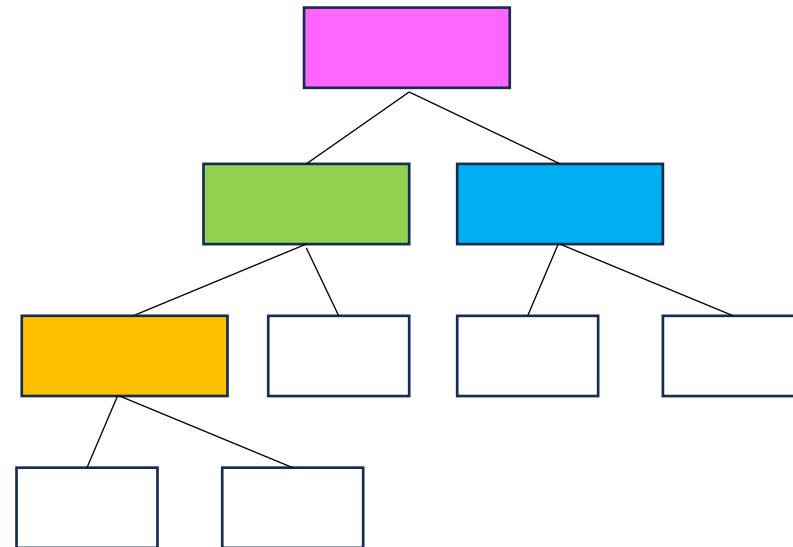
Bootstrapping is used as random sampling with replacement.



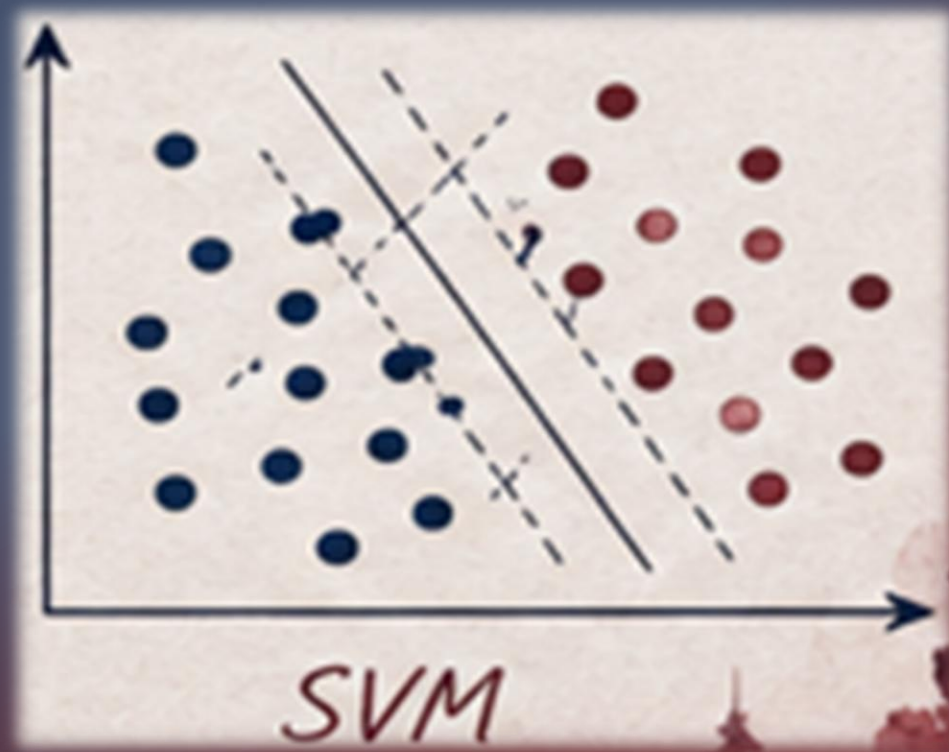


4. Out-of-Bag Error:

Some samples may never be selected in the bootstrap samples; these are used for evaluation.



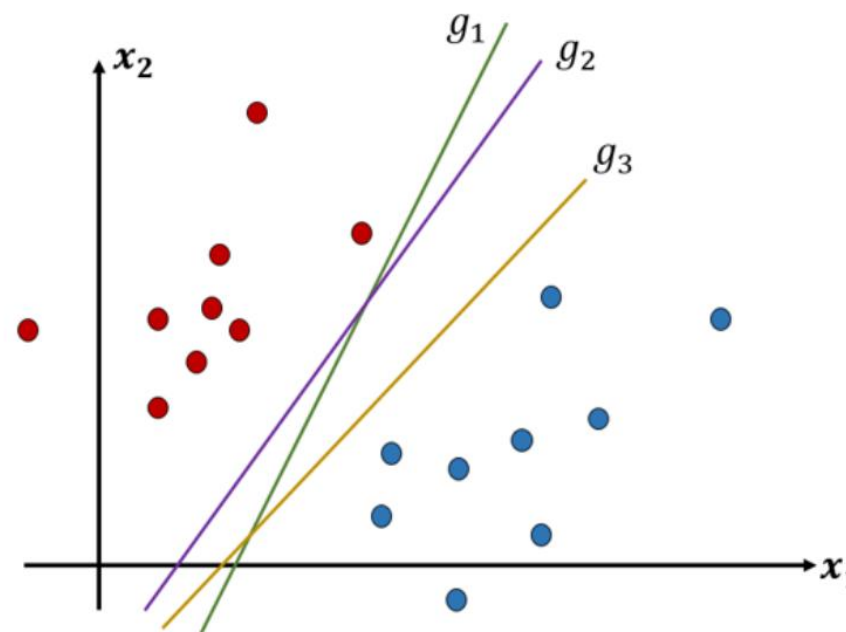
3



Separable Classes

- ❑ Let $x_i (i \in \{1, 2, 3, \dots, n\})$ be the feature vectors of the training set X
- ❑ This data belong to 2 classes y_1 and y_2
- ❑ The goal, once more, is to design a decision boundary:

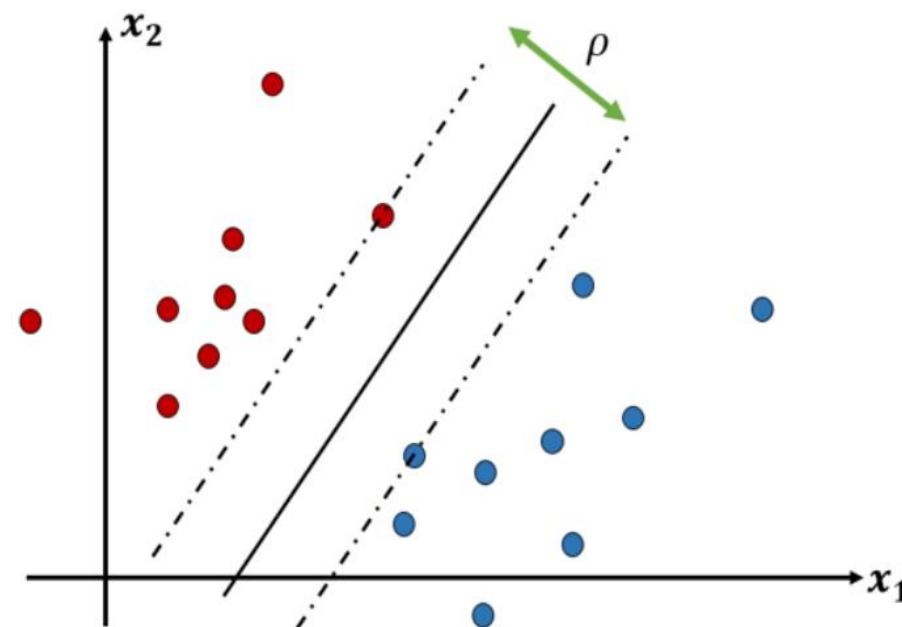
$$g(x) = w^T x + b$$



Generalization performance

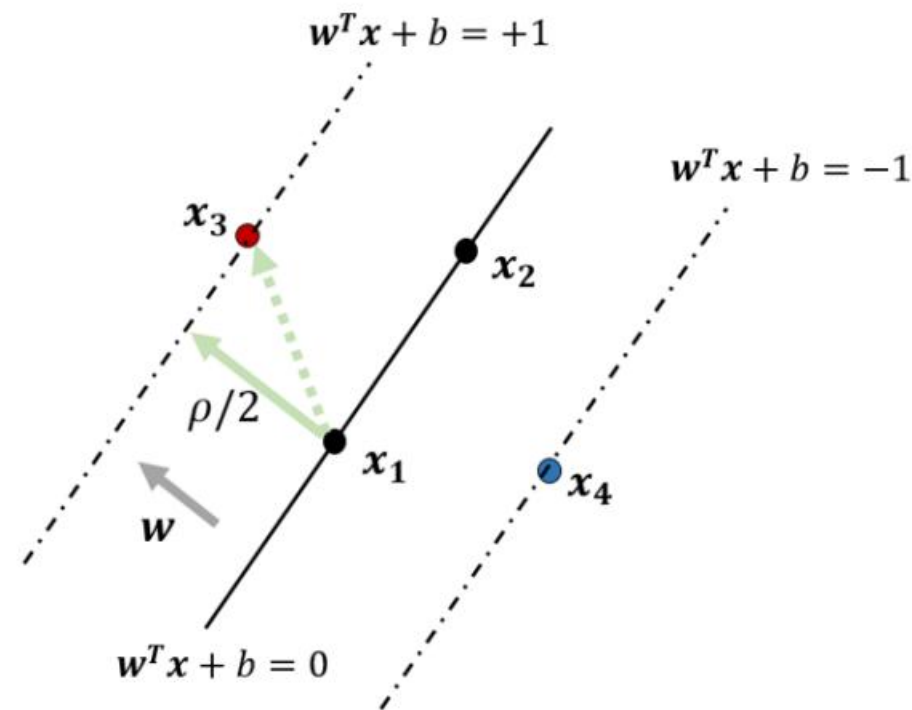
- capability of the classifier, designed using the training data set, to operate satisfactorily with data outside this set.

$$\begin{cases} w^T x + b \geq 1 & \text{if } y_1 \\ w^T x + b \leq -1 & \text{if } y_2 \end{cases}$$

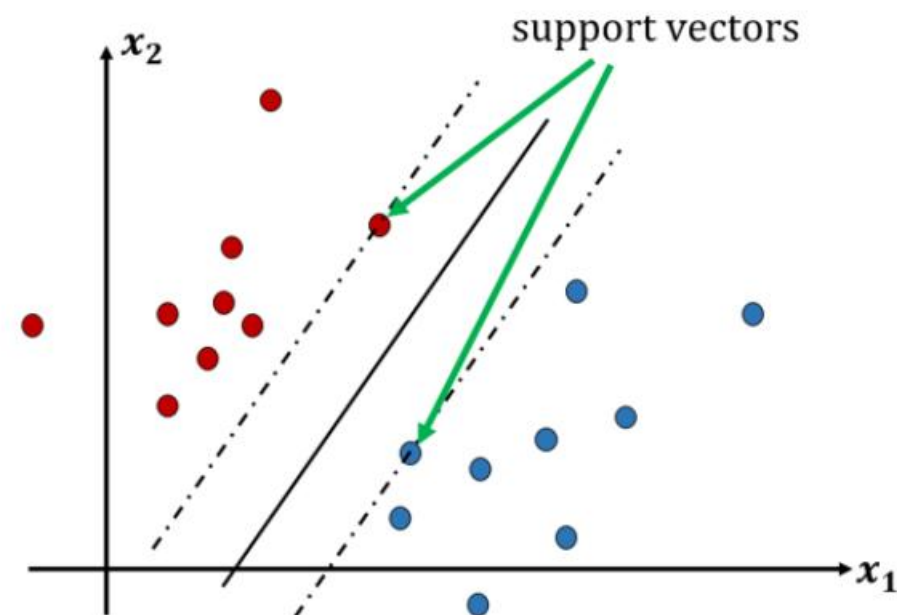


consume: $g(x_1) = g(x_2) = 0$

$$\frac{1}{2}\rho = \frac{\mathbf{w}^T(\mathbf{x}_3 - \mathbf{x}_1)}{\|\mathbf{w}\|} = \frac{(\mathbf{w}^T \mathbf{x}_3 + b) - (\mathbf{w}^T \mathbf{x}_1 + b)}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$$



- Samples farther away from the margin have little influence on the boundary
- The optimal decision boundary of an SVM is unique. Intuitively, we can see that there cannot exist multiple decision boundaries achieving the same maximum margin.
- SVM is robust to data outliers, as vectors farther away from the margin of the classes do not influence the model.
- SVM is also little affected by data distribution and density.

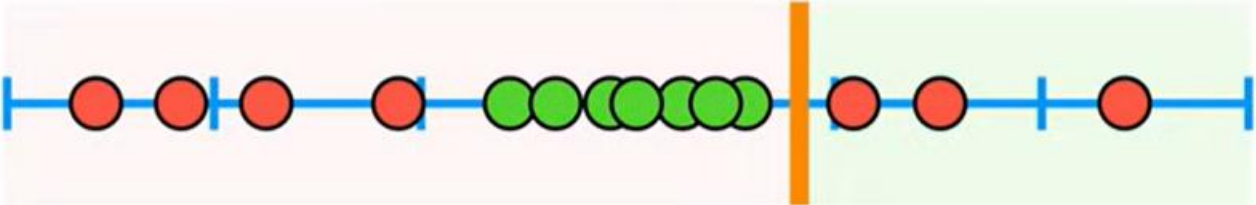


How to find best MARGINE in this example?

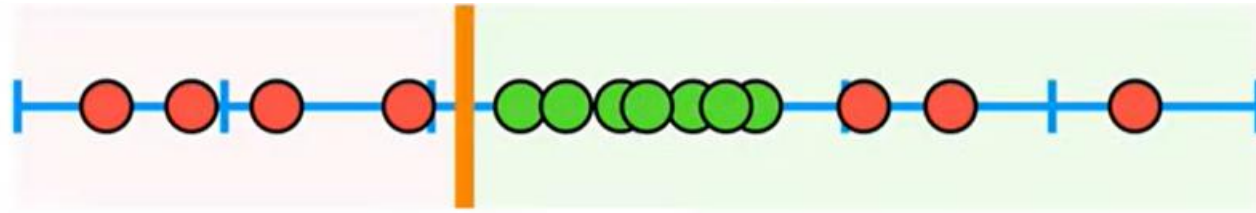
Dosage (mg):



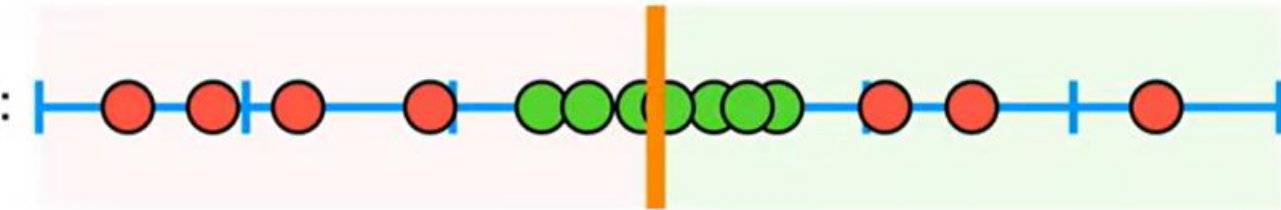
Dosage (mg):



Dosage (mg):



Dosage (mg):





SVM Can help ud in this example. How?

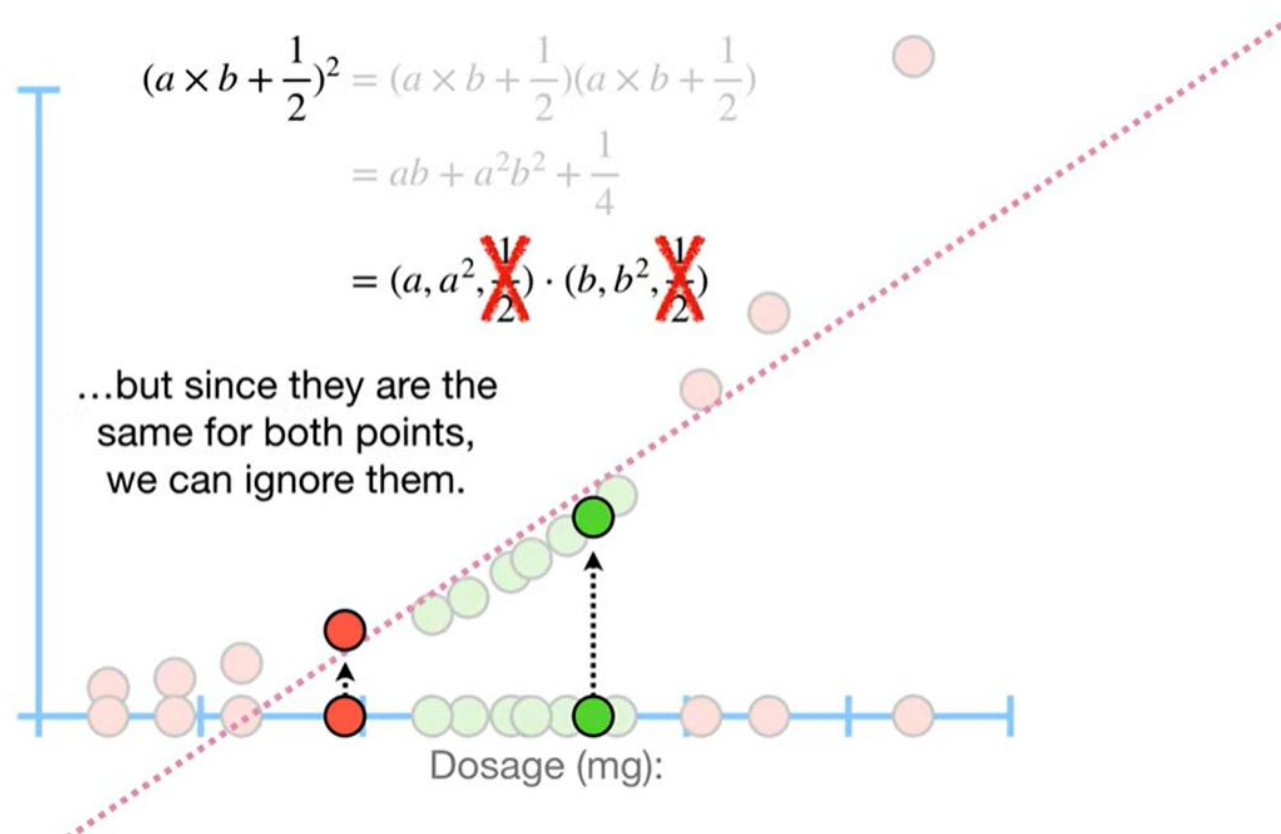
- SVMs were developed in the 1990s by Vladimir N. Vapnik and his colleagues, and they published this work in a paper titled "Support Vector Method for Function Approximation, Regression Estimation, and Signal Processing"¹ in 1995.
- The “kernel trick” helps to reduce some of that complexity, making the computation more efficient, and it does this by replacing dot product calculations with an equivalent kernel function.
- kernel functions include:
 - Polynomial kernel
 - Radial basis function kernel (also known as a Gaussian or RBF kernel)

Polynomial kernel

$$(a \times b + r)^d$$

- **a** and **b** refer to two different observation in the dataset
- **r** determines the coefficient of the polynomial
- **d** is degree of the polynomial

Example set $r = 1/2$ and $d = 2$





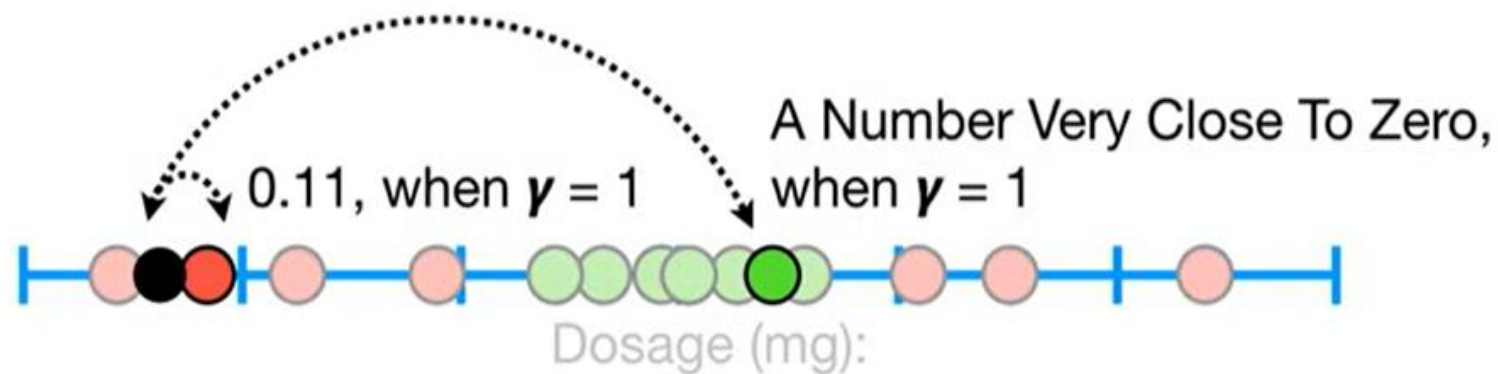
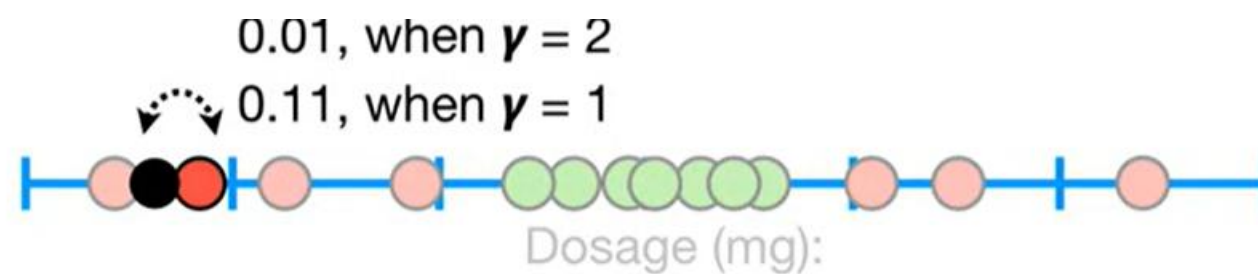
Radial basis function kernel

- When you have no prior knowledge about the distribution or geometric structure of your dataset, RBF is typically the safest and most powerful default starting option.
- It works beautifully when you have a reasonable number of training samples but highly complex relationship patterns between your features.

- Mathematical Formula:

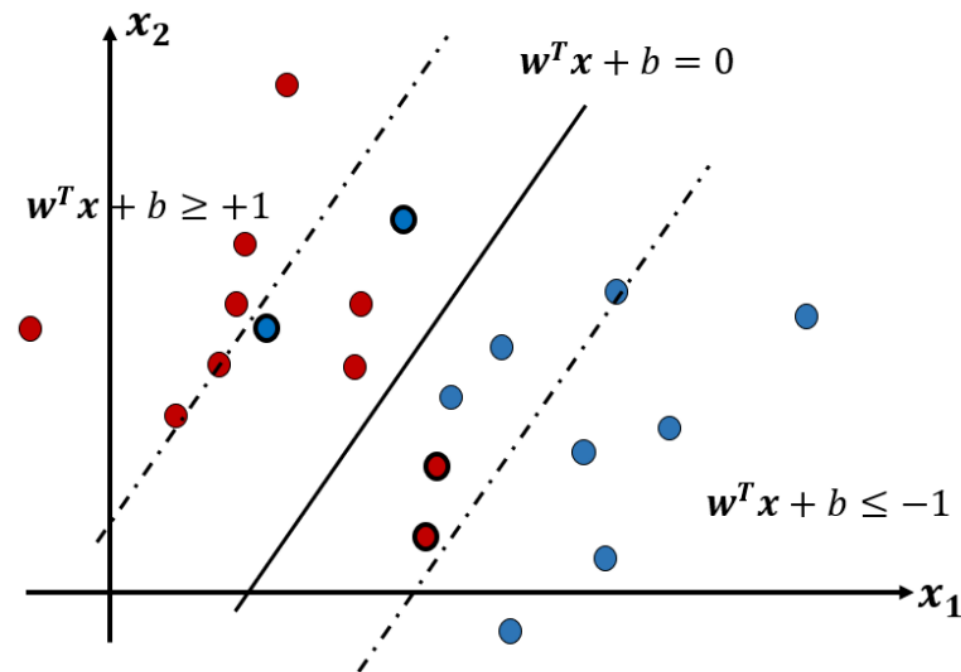
$$e^{-\gamma(a-b)^2}$$

- If Gamma it's too low, the model might underfit because the boundary becomes too rigid to capture complex patterns.
- If Gamma its top high This easily leads to overfitting. The model gets a perfect score on the training data but fails completely on unseen test data.



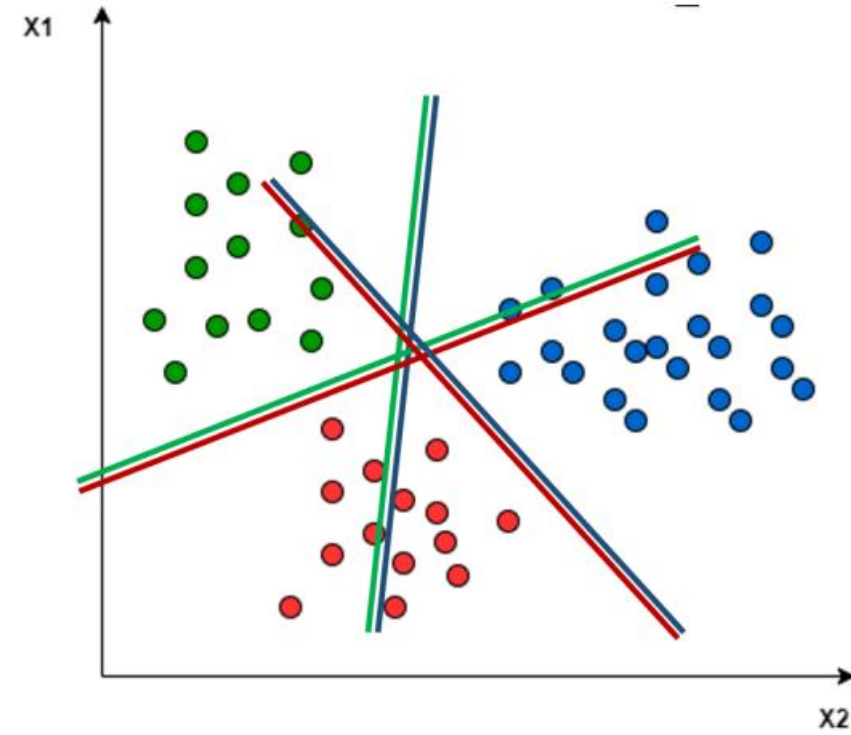
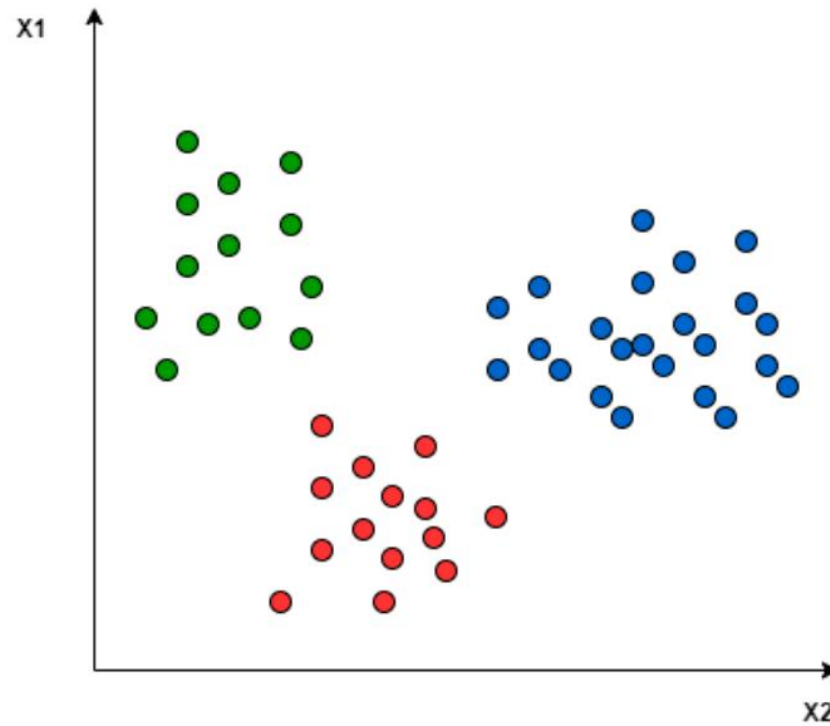
SVM= kernel + sv

$$g(\mathbf{x}) = \sum_{i \in \text{SVs}} \lambda_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$$



Generalization to multi-class classification

$$\frac{n(n-1)}{2} \text{ SVMs.}$$



Thanks for your
attention!